Stat 286: Causal Inference with Applications, Fall 2025

Section 11: Panel Data

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1 Difference-in-Differences

1.1 Motivation

So far, we have only considered settings in which each unit is either treated or not treated once, and we observe a single outcome. In many applications, however, we observe the *same* units repeatedly over time. In this section we focus on the case where all units start in control and, at some later time, a subset of units is exposed to the treatment. Our parameter of interest is the average treatment effect on the treated (ATT), since we care about the effect for those units that actually receive the treatment.

Because time now plays a central role, we will use t to index time and reserve $Z_{it} \in \{0, 1\}$ for the treatment indicator (rather than T, which we used earlier).

How should we conduct causal inference in this setting? Consider the simple case where treatment is first given at time $t+1$, and no unit is treated at time t. For a treated unit i, the causal effect at time $t+1$ is

$$\tau_{it+1} = Y_{it+1}(1) - Y_{it+1}(0),$$

but we only observe either $Y_{it+1} = Y_{it+1}(1)$ and $Y_{it} = Y_{it}(0)$, but not $Y_{it+1}(0)$.

Thus, a first idea is to compare the same unit *before* and *after* treatment,

$$Y_{it+1} - Y_{it}.$$

This is illustrated in the left panel of Figure 1. However, for a treated unit i, this implicitly assumes $Y_{it}(0) = Y_{it+1}(0)$, i.e., that the untreated outcome would not change over time. This is often unrealistic, as it fails to account for changes over time.

A second idea is to use units that remain in the control group, as in the right panel of Figure 1. We could compare the treated unit i to some control unit j at time $t+1$,

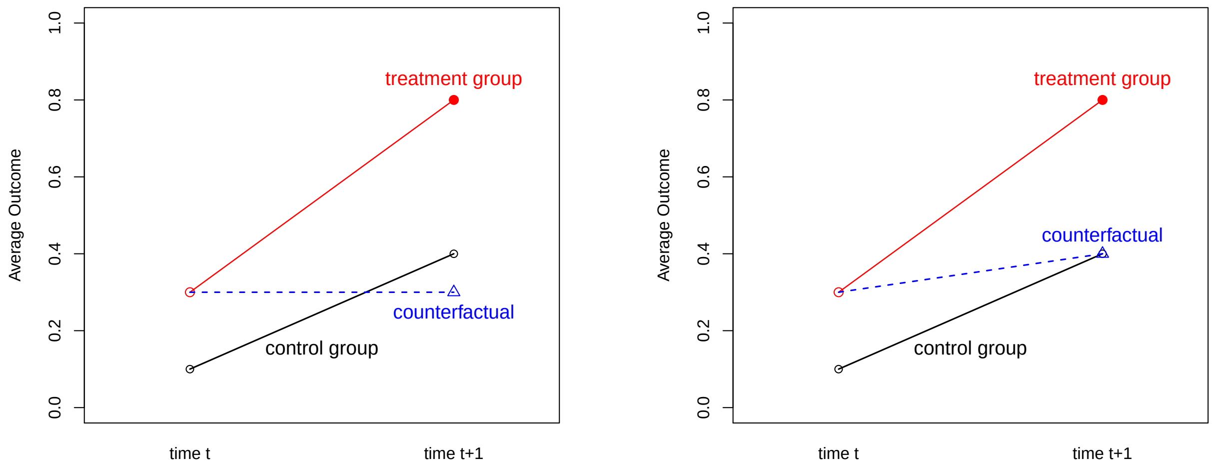
$$Y_{it+1}(1) - Y_{jt+1}(0).$$

Now we are comparing different units at the same time, which assumes $Y_{it+1}(0) = Y_{jt+1}(0)$. This is also hard to justify, as it fails to account for systematic difference between the treated and control (recall that treatment is not randomized).

In what follows, we develop a method called *difference-in-differences* that aims to estimate $Y_{it+1}(1) - Y_{it+1}(0)$ (more precisely, its average) more credibly by combining the time comparison and the cross-sectional comparison. The key assumption is that units remaining in control satisfy the *parallel trends* assumption, illustrated in Figure 2.

1.2 Running example: Minimum wage and employment.

To fix ideas, consider the classic minimum wage study of Card and Krueger [1993]. We observe fast-food restaurants in two neighboring states, New Jersey and Pennsylvania, at two points in time: before ($t = 0$) and after ($t = 1$) New Jersey raises its minimum wage. Let Y_{it} be employment at restaurant i at time t, and let $Z_{it} = 1$ if restaurant i is in New Jersey at time $t = 1$ (after the reform), and $Z_{it} = 0$ otherwise. All restaurants are untreated at $t = 0$, and only New Jersey restaurants become treated at $t = 1$, while Pennsylvania restaurants remain in the control group throughout. Our goal is to learn the effect of the minimum wage increase on employment in treated restaurants, that is, the ATT in this two-by-two setting.



Before-After Cross-Sectional

Figure 1: Taken from Kosuke Imai's lecture slides.

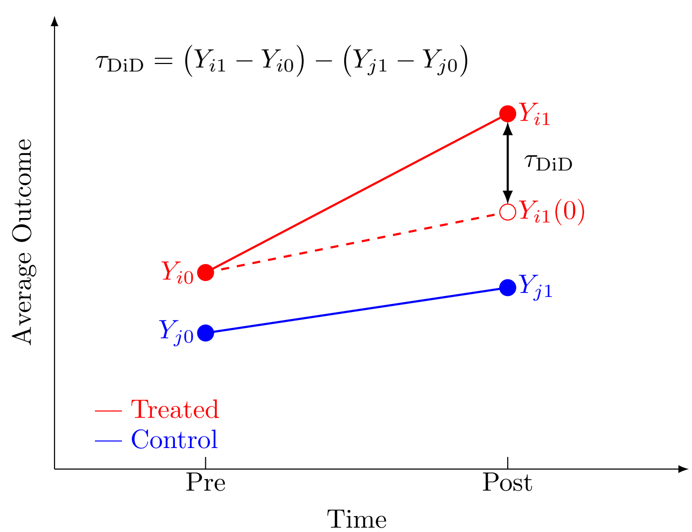


Figure 2: Difference-in-differences in the 2×2 case. Counterfactual is imputed by adding the trend of the control to Y_{i0}.

1.3 Setup

We focus on the simple 2×2 case (two time periods and treatment groups), where we have

- Two time periods: $t = 0$ (pre-treatment), $t = 1$ (post-treatment)
- G_i: treatment group indicator (time-invariant)
- $Z_{it} = tG_{it}$: treatment assignment indicator
- Potential Outcomes: $Y_{it}(z)$ for $t = \{0, 1\}$ and $z \in \{0, 1\}$
- Observed Outcome: $Y_{it} = Y_{it}(Z_i)$.

Our estimand of interest is the ATT,

$$\tau_{\text{ATT}} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1].$$

As discussed in the motivation section, for $G_i = 1$, $Y_{i1}(1) = Y_{i1}$ is observed, but we do not observe $Y_{i1}(0)$. To make identification possible, we make the following assumption.

Assumption 1 (Parallel Trend)**.** *Non-adopters evolve in parallel regardless of group membership, i.e.,*

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0].$$

Assumption 1 is illustrated in Figure 2. It combines the before/after comparison with the cross-sectional comparison discussed in the motivation section. Under Assumption 1 we can identify the ATT

$$\begin{aligned}
\tau_{\text{ATT}} &= \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] \\
&= \mathbb{E}[Y_{i1}(1) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) \mid G_i = 1] \\
&= \mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) \mid G_i = 1] \\
&= \mathbb{E}[Y_{i1} \mid G_i = 1] - (\mathbb{E}[Y_{i0}(0) \mid G_i = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]) \qquad \because \text{(parallel trend)} \\
&= \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{\text{after treated}} - (\underbrace{\mathbb{E}[Y_{i0} \mid G_i = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0]}_{\text{before treated + trend of control}}) \\
&= \underbrace{\mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1]}_{\text{before/after treated}} - \underbrace{\mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0]}_{\text{before/after control}}.
\end{aligned}$$

As we can see, under the parallel trend assumption, the ATT imputes

$$\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i0} \mid G_i = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0],$$

and ends up being an adjusted before/after comparison of the treated group with the trend in control.

This suggests the following non-parametric estimator

$$\hat{\tau}_{\text{ATT}}^{\text{DiD}} = \frac{1}{n_1}\sum_{i=1}^{n} G_i(Y_{i1} - Y_{i0}) - \frac{1}{n_0}\sum_{i=1}^{n}(1 - G_i)(Y_{i1} - Y_{i0})$$

where $n_1 = \sum_{i=1}^{n} G_i$ and $n_0 = n - n_1$. By the above calculations $\mathbb{E}[\hat{\tau}_{\text{ATT}}^{\text{DiD}}] = \tau_{\text{ATT}}$. Note that instead of comparing outcomes, we compare trends of outcomes.

In our above calculations we have implicitly used the following assumption.

Assumption 2 (No Anticipation)**.** *Future treatment has no causal effect on current outcomes. Formally, for any unit i and time t such that $Z_{is} = 0$ for all $s \leq t$ (i.e. unit i has not yet been treated by time t),*

$$Y_{it} = Y_{it}(0).$$

This assumption is violated if units anticipate future treatment and adjust their behavior in advance. For example, if people expect an increase in interest rates, they may choose to take out loans earlier.

1.4 Linear Model for the Difference-in-Differences

A sufficient condition for Assumption 1 is that the untreated potential outcomes have a separable unit–time structure:

$$\mathbb{E}[Y_{it}(0)] = \alpha_i + \beta_t, \tag{1}$$

with α_i capturing time-invariant heterogeneity across units and β_t capturing shocks common to all units at time t. In the 2×2 case we can normalize $\beta_0 = 0$ and write $\beta_1 = \beta$.

Using Equation (1) together with Assumption 2, we obtain the following structure for *untreated* observations (units and times that have not yet received treatment):

$$Y_{it} = Y_{it}(0) = \alpha_i + \beta_t + u_{it}(0),$$

where $u_{it}(0) := Y_{it}(0) - \mathbb{E}[Y_{it}(0)]$ is a zero-mean noise term with $\mathbb{E}[u_{it}(0) \mid \alpha_i, \beta_t] = 0$ (note that this also implies that $\mathbb{E}[u_{it}(0) \mid G_i] = 0$ since α_i captures all units effects and G_i is a unit effect). In general we can write

$$Y_{it} = Y_{it}(0) + Z_{it}\big(Y_{it}(1) - Y_{it}(0)\big) = Y_{it}(0) + Z_{it}\,\tau_{it},$$

where $\tau_{it} := Y_{it}(1) - Y_{it}(0)$ is the (possibly heterogeneous) treatment effect. We decompose $\tau_{it} = \tau_{\text{ATT}} + v_{it}$, with $\mathbb{E}[v_{it} \mid Z_{it} = 1, \alpha_i, \beta_t] = 0$. Substituting this into the expression for Y_{it} and using (1) gives

$$Y_{it} = \alpha_i + \beta_t + \tau_{\text{ATT}} Z_{it} + \varepsilon_{it}, \tag{2}$$

where $\varepsilon_{it} := u_{it}(0) + Z_{it} v_{it}$ satisfies $\mathbb{E}[\varepsilon_{it} \mid \alpha_i, \beta_t, Z_{it}] = 0$. It turns our that if we run this *two-way-fixed effect* regression in Equation 2, the the estimate $\tau^{\text{FE}}_{\text{ATT}}$ on τ_{ATT} is the same for the non-parametric DiD estimator, i.e., $\hat{\tau}^{\text{FE}}_{\text{ATT}} = \hat{\tau}^{\text{DiD}}_{\text{ATT}}$. In this sense, the parallel trend Assumption 1 is equivalent to the structural assumption in Equation 1. This only holds in the 2×2 case.

1.5 Time Invariant vs Time Varying Confounders

DiD naturally adjusts for *time-invariant* confounders, as can be seen from Equation (1), where α_i captures all unit-specific, time-invariant heterogeneity. However, it does not automatically adjust for *time-varying* confounders whose evolution differs between treated and control groups.

To see this, suppose the untreated potential outcomes follow

$$Y_{it}(z) = \alpha_i + \beta_t + \delta_t G_i + \tau z + \eta_{it},$$

where $G_i \in \{0, 1\}$ indicates the treated group and δ_t is a group-specific time trend. Then $\tau_{\text{ATT}} = \tau$ as before, but our parallel trend identification strategy yields

$$\begin{aligned}\mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_i = 0] &= \mathbb{E}\big[Y_{i1}(1) - Y_{i0}(0) \mid G_i = 1\big] - \mathbb{E}\big[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0\big] \\ &= \tau + \big(\delta_1 - \delta_0\big).\end{aligned}$$

If δ_t is constant over time, this bias term disappears and DiD recovers τ. Otherwise, DiD is biased.

1.6 Variance Estimation

Variance can be estimated in (at least) the following ways:

- **Cluster-robust standard errors**: Using the fixed-effects model, calculate cluster-robust standard errors, with clustering at the unit level.
- **Cluster bootstrap**: Once you sample the unit, you must sample all its time steps.

1.7 Diagnostics and Validation

- **Pre-treatment plots**: For $g \in \{0, 1\}$ and $t = 1, \dots, T$ (requires multiple time steps), compute

$$\bar{Y}_{gt} = \frac{1}{n_g} \sum_{i:G_i=g} Y_{it}$$

and plot $\bar{Y}_{0t}$ and $\bar{Y}_{1t}$ over pre-treatment periods. Approximately parallel lines suggest plausibility of parallel trends. (This generalizes to multiple groups.)

- **Event-study pre-trend tests**: With multiple pre-periods and staggered adoption times H_i, estimate

$$Y_{it} = \alpha_i + \beta_t + \sum_{k \neq -1} \delta_k D_{it}^k + \varepsilon_{it},$$

where $D_{it}^k = \mathbf{1}\{G_i = 1,\ t - H_i = k\}$ is a relative-time indicator and $k = -1$ is the omitted (baseline) period. A pre-trend test is the joint test

$$H_0 : \ \delta_k = 0 \quad \text{for all } k < 0,$$

e.g. via an F-test. Failure to reject is consistent with parallel pre-trends.

- **Placebo test**: Choose a pseudo-treatment date $H_i^{\text{pl}} < H_i$ and construct a placebo treatment path $Z_{it}^{\text{pl}} = \mathbf{1}\{t > H_i^{\text{pl}}\}G_i$ that lies entirely in the pre-treatment period. Re-estimate the DiD/TWFE model using Z_{it}^{pl} in place of Z_{it}. Under parallel trends, the corresponding placebo effect $\hat{\tau}^{\text{pl}}$ should be close to zero.

1.8 Extensions

- **Lagged Outcome Model**: Another approach to panel data is to treat the pre-treatment outcome as a covariate. Consider the linear lagged outcome model

$$Y_{i1}(z) = \alpha + \rho Y_{i0} + \tau z + \varepsilon_{i1}(z), \tag{3}$$

Identification of τ relies on

$$\{Y_{i1}(1), Y_{i1}(0)\} \perp Z_{it} \mid Y_{i0}. \tag{4}$$

Compared to DiD, this approach explicitly adjusts for the possibility that Y_{i0} both predicts treatment and affects future outcome. Its drawback is that it requires a randomization-type assumption conditional on Y_{i0}. It is neither stronger or weaker as the parallel trend Assumption 1.

- **Multiple Time Periods and Groups**: Everything discussed so far can be extended beyond the 2×2 case. For reference we refer to Wager [2024, Chapter 13], to the recent paper Borusyak et al. [2024] or to Callaway and Sant'Anna [2021], Imai et al. [2023] (below).

- **Covariates and Doubly-Robust Estimation**: Following Callaway and Sant'Anna [2021], we can indluce baseline covariates X_i and assume *conditional* parallel trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid X_i = x, G_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid X_i = x, G_i = 0].$$

Let $\pi(x) = \Pr(G_i = 1 \mid X_i = x)$ be the propensity score and $m(x) = \mathbb{E}(Y_{i1} - Y_{i0} \mid G_i = 0, X_i = x)$ the outcome regression for controls. A doubly-robust ATT estimator is then

$$\hat{\tau}_{\text{ATT}}^{\text{DR}} = \frac{1}{n_1} \sum_{i=1}^{n} \Big(Y_{i1} - Y_{i0} - \hat{m}(X_i) \Big) \frac{G_i - \hat{\pi}(X_i)}{1 - \hat{\pi}(X_i)}.$$

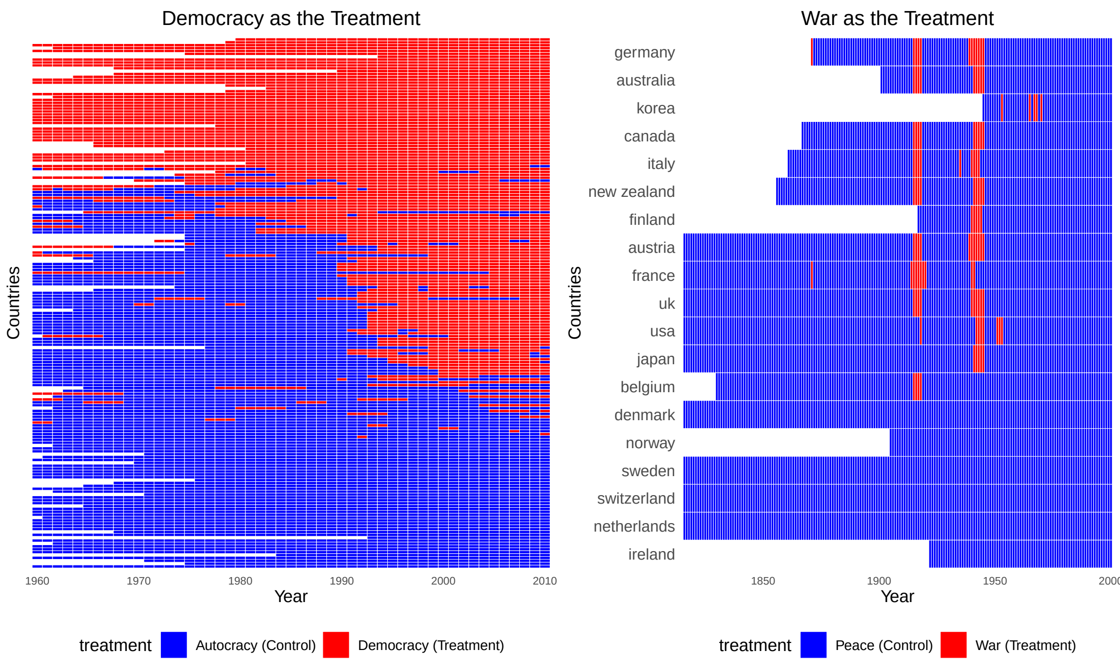
This can also be extended to multiple time periods in a staggered adoption design:

  - never-treated group: $G_i = 0$

- K treated groups (sorted by timing): $G_i = 1, 2, \dots, K$
- Estimate ATT for each treated group and then aggregate:

$$\widehat{\text{ATT}} = \sum_{g=1}^{K} \Pr(G_i = g)\, \widehat{\text{ATT}}(g)$$

- **Panel Match**: This is taken from Kosuke Imai's slides with corresponding paper Imai et al. [2023]. How do deal with repeated treatments over time?



  - Choose number of lags L for confounder adjustment.
  - Choose number of leads F for short- or long-term effects.

Define

$$\text{ATT} = \mathbb{E}\Big[Y_{i,t+F} \;\Big|\; Z_{it} = 1,\ Z_{i,t-1} = 0,\ \{Z_{i,t-\ell}\}_{\ell=2}^{L}\Big] - \mathbb{E}\Big[Y_{i,t+F} \;\Big|\; Z_{it} = 0,\ Z_{i,t-1} = 0,\ \{Z_{i,t-\ell}\}_{\ell=2}^{L}\Big].$$

Procedure:

1. Find treated observations with $Z_{i,t-1} = 0$ and $Z_{it} = 1$.
2. For each treated (i, t), form a matched set M_{it} of control observations with identical treatment history from $t-1$ to $t-L$.
3. Refine matched sets via classical matching/weighting methods to adjust time-invariant and time-varying confounders.
4. Compute the DiD estimator

$$\widehat{\text{ATT}}_{\text{PM}} = \frac{1}{\sum_{i=1}^{N} \sum_{t=L+1}^{T-F} D_{it}} \sum_{i=1}^{N} \sum_{t=L+1}^{T-F} D_{it} \left[(Y_{i,t+F} - Y_{i,t-1}) - \sum_{i' \in M_{it}} w_{i'it} (Y_{i',t+F} - Y_{i',t-1}) \right],$$

where $D_{it} = \mathbf{1}\{Z_{i,t-1} = 0, Z_{it} = 1\}$.

1.9 When to use DiD

- Appropriate when you have many treated and untreated units and treatment occurs at a common time (or staggered times).
- When multiple pre-treatment periods are available to check pre-trends and perform placebo tests.
- When unobserved confounding is believed to be largely time-invariant.
- When treatment anticipation is unlikely.

2 Synthetic Control

2.1 Motivation

As we have seen above, the DiD approach combines the before/after and cross-sectional comparisons by assuming parallel trends in the untreated potential outcomes to impute the missing $Y_{i,t+1}(0)$ for treated units. However, the plausibility of this parallel trends assumption is often questionable.

The synthetic control method offers an alternative. First proposed by Abadie and Gardeazabal [2003], Abadie et al. [2010], it replaces parallel trends with the assumption that the treated unit's counterfactual path lies in the *convex hull* of the control units. Concretely, we posit weights $w_j \geq 0$ with $\sum_j w_j = 1$ such that, for all pre-treatment periods $s \leq t$,

$$Y_{is}(0) \approx \sum_j w_j Y_{js}(0),$$

and then construct the counterfactual at $t+1$ as

$$\hat{Y}_{i,t+1}(0) = \sum_j w_j Y_{j,t+1}.$$

Thus, synthetic control "builds" a counterfactual for the treated unit as a weighted average of control units that best reproduces its pre-treatment trajectory. We fit the potential outcome directly and don't rely on averages. This only works well with a large pool of untreated units and one (or few) treated units, as well as a long pre-treatment trajectory for accurate predictions.

2.2 Setup

Consider the setting with N units and T time periods. We assume only one treated unit $i = N$ receiving treatment at time T (this can be relaxed to multiple units and multiple post-treatment periods). We observe

- $t = 1, \ldots, T-1$: $Y_{it} = Y_{it}(0)$,
- $t = T$: $Y_{it} = Y_{iT}(0)$ for $i < N$ and $Y_{NT} = Y_{iT}(1)$.

Our causal estimand of interest is the individual effect of the treated unit

$$\tau_N = Y_{NT}(1) - Y_{NT}(0) = Y_{NT} - Y_{NT}(0).$$

The setup is illustrated in Figure 3.

2.3 Convex Hull

As with DiD, we would like to impute the missing value for $Y_{NT}(0)$. To this end, we make the following assumption.

Assumption 3 (Convex Hull)**.** *There exist weights $(w_1, \ldots, w_{N-1})$ with $w_i \geq 0$ and $\sum_{i=1}^{N-1} w_i = 1$ such that the untreated potential outcomes of the treated unit N lie in the convex hull of the control units, i.e.,*

$$Y_{Nt}(0) = \sum_{i=1}^{N-1} w_i \, Y_{it}(0), \qquad t = 1, \ldots, T.$$

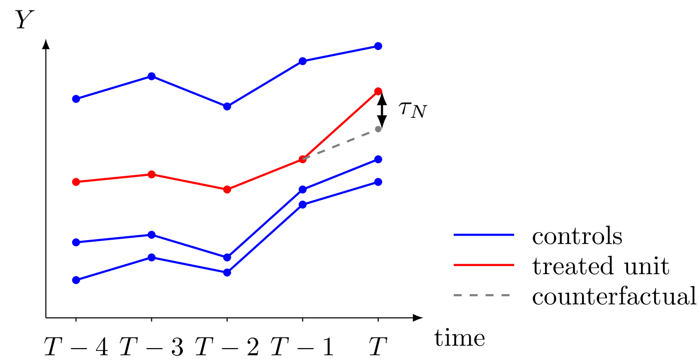


Figure 3

Assumption 3 states that $Y_{Nt}(0)$ is a *time-invariant* weighted average of $\{Y_{it}(0)\}_{i=1}^{N-1}$: the same weight vector $w = (w_1, \ldots, w_{N-1})$ links unit N to the donor units in every period. This encodes a stable structural relationship between the treated unit and the controls over time.

The constraints $w_i \geq 0$ and $\sum_{i=1}^{N-1} w_i = 1$ restrict the synthetic control to be an *interpolation* of existing control units rather than an extrapolation. In particular, the synthetic path $\sum_{i=1}^{N-1} w_i Y_{it}(0)$ remains inside the convex hull of the control trajectories; see Figure 4. As synthetic control is foremost a prediction method (we predict a single unit, compared to averages), avoiding extrapolation is particularly important (recall OLS). For example, if all control regions have unemployment rates between 5% and 10%, an unconstrained regression could fit the pre-treatment data using large (negative) weights and predict a counterfactual of 20%, while the convex-combination restriction forces the synthetic control to remain in the plausible range $[5\%, 10\%]$.

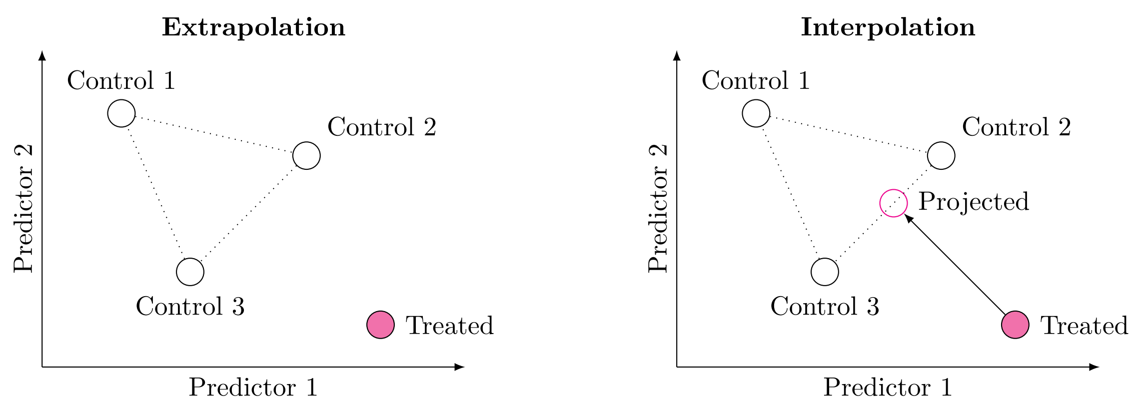


Figure 4: Extrapolation (left) predicts the treatment. Interpolation (right) predicts the treatment and projects it to the convex hull of controls.

In practice, we fit

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} w_i Y_{it} \right)^2 \quad \text{subject to} \quad \sum_{i=1}^{N-1} w_i = 1,\ w_i \geq 0\ \forall i, \tag{5}$$

and estimate $\widehat{Y}_{NT}(0) = \sum_{i=1}^{N-1} \hat{w}_i Y_{iT}$. We then set

$$\hat{\tau}_N = Y_{NT} - \widehat{Y}_{NT}(0).$$

To obtain a good fit this usually requires long, stable pre-treatment trajectories.

Remark 1. Throughout this section we implicitly assume *no spillover effects*: treatment of unit N does not affect the outcomes of the control units. This assumption is particularly important for synthetic control, since the control units are used to construct the counterfactual trajectory for the treated unit over many periods. If treatment spills over to the donor pool, the synthetic control will no longer represent an untreated path.

2.4 Model-Based Justifications

Assumption 3 may look intuitive, but it might be hard to justify. Abadie et al. [2010] provide two models that satisfy the convex hull assumption

- **Factor Model**:
$$Y_{it}(0) = \gamma_t + \delta_t^\top X_i + \xi_t^\top U_i + \varepsilon_{it},$$
  - X_i are observed unit-level covariates,
  - U_i are unobserved time-invariant factors,
  - Generalization of the linear two-way fixed effects model,
  - Assumption: exist weights $(w_1, \dots, w_{N-1})$ with $w_i \geq 0$, $\sum_{i=1}^{N-1} w_i = 1$ such that $\sum_{i=1}^{N-1} w_i X_i = X_N, \sum_{i=1}^{N-1} w_i U_i = U_N$.
- **Autoregressive Model**:
$$\begin{aligned} Y_{it}(0) &= \rho_t Y_{i,t-1}(0) + \delta_t^\top X_{it} + \varepsilon_{it}, \\ X_{it} &= \lambda_{t-1} Y_{i,t-1}(0) + \Delta_{t-1} X_{i,t-1} + \nu_{it}, \end{aligned}$$
  - Time-varying covariates,
  - Past outcomes can affect current treatment.
  - No unobserved time-invariant confounders.

2.5 Inference

We discuss two methods for inference.

2.5.1 Placebo Tests

Our aim is to test $H_0 : Y_{NT}(1) = Y_{NT}(0)$ and derive a confidence interval. To this end, we construct placebo effects for each control unit $j = 1, \dots, N-1$:

- Pick a unit $j < N$ and treat it as the treated unit,
- Fit the synthetic control estimate to obtain $\widehat{Y}_{jt}(0)$ (include treated unit $i = N$),
- Calculate $\hat{\tau}_N = Y_{jT} - \widehat{Y}_{jT}(0)$.

We do this for every unit $j = 1, \dots, N$. Assuming that $\{\hat{\tau}_j\}_{j=1}^N$ are exchangeable, we can calculate the (two-sided) permutation p-value

$$p = \frac{1}{(N-1)+1}\left(1 + \sum_{j=1}^{N-1} \mathbf{1}\big\{|\hat{\tau}_j| \geq |\hat{\tau}_N|\big\}\right).$$

Remark 2. Assuming that $\{\hat{\tau}_j\}_{j=1}^N$ are *exchangeable* means that, under $H_0 : Y_{NT}(1) = Y_{NT}(0)$, the joint distribution of the effects does not depend on which unit is labeled as "treated". Formally, for any permutation π of $\{1, \dots, N\}$,

$$(\hat{\tau}_1, \dots, \hat{\tau}_N) \overset{d}{=} (\hat{\tau}_{\pi(1)}, \dots, \hat{\tau}_{\pi(N)}).$$

It also means that the marginals are the same (identically distributed), which can be questionable.

As always, once we know how to calculate p-values, we can invert them to obtain a confidence interval. Let $H_0 : Y_{NT}(1) = Y_{NT}(0) + \tau_0$ and set

$$\hat{\tau}_N(\tau_0) = (Y_{NT} - \tau_0) - \widehat{Y}_{NT}(0).$$

The confidence interval will then consist of all τ_0 for which we don't reject the null. Note that this method of inference requires a large N to be powerful.

2.5.2 Variance

One can assume that $Y_{NT} = Y_{NT}(1)$ is fixed and the only uncertainty arises from the estimation of $Y_{NT}(0)$, using noisy control trajectories [Doudchenko and Imbens, 2016]. A simple design-based variance estimator treats the placebo effects as draws from the baseline noise distribution and sets

$$\widehat{\text{Var}}(\hat{\tau}_N) = \frac{1}{N-1} \sum_{j=1}^{N-1} \hat{\tau}_j^2.$$

2.5.3 Diagnostics

- Look at pre-treatment fit of treated and placebo units.
- Compare the RMSE before and after treatment for placebo units.

2.6 Extensions

- **Adding Covariates**: Define $Z_i = \begin{pmatrix} Y_i \\ X_i \end{pmatrix}$, $\quad Y_i = (Y_{i1}, \ldots, Y_{i,T-1})^\top$, $\quad X_i = (X_{i1}^\top, \ldots, X_{i,T-1}^\top)^\top$ and solve

$$\hat{w} = \arg\min_w \Big(Z_N - \sum_{i=1}^{N-1} w_i Z_i \Big)^\top \hat{\Sigma}_Z^{-1} \Big(Z_N - \sum_{i=1}^{N-1} w_i Z_i \Big)$$
$$\text{subject to} \qquad \sum_{i=1}^{N-1} w_i = 1, \qquad w_i \geq 0 \text{ for } i = 1, \ldots, N-1,$$

  where $\hat{\Sigma}_Z$ is a covariance (or weighting) matrix for the components of Z_i.

- **Bias Correction**: Similar to the bias-corrected matching estimator, we can adjust for possible bias due to poor pre-treatment fit, see Ben-Michael et al. [2021].
- **Regularization**: Standard synthetic control method tends to overfit. Abadie and L'hour [2021] propose additional regularization.

$$\arg\min_w \left\{ \underbrace{\sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} w_i Y_{it} \right)^2}_{\text{component-wise fit}} + \lambda \underbrace{\sum_{i=1}^{N-1} w_i \sum_{t=1}^{T-1} \big(Y_{Nt} - Y_{it} \big)^2}_{\text{aggregate fit}} \right\}$$

  - $\lambda \to 0$: synthetic control.
  - $\lambda \to \infty$: nearest neighbor matching.

- **Staggered Adoption**: Synthetic control can be extended to staggered adoption, see Ben-Michael et al. [2022].

- **Matrix Completion**: Synthetic control is essentially just a restricted weighted regression where we are interest in filling in the missing data in the following matrix (assuming T_1 post-time periods and N_1 treated units).

$$\mathbf{Y}(0) = \left[\begin{array}{ccc|ccc} Y_{1,1} & \cdots & Y_{1,T-1} & Y_{1,T} & \cdots & Y_{1,T+T_1} \\ \vdots & & \vdots & \vdots & & \vdots \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ Y_{N,1} & \cdots & Y_{N,T-1} & Y_{N,T} & \cdots & Y_{N,T+T_1} \\ \vdots & & \vdots & \vdots & & \vdots \\ Y_{N+N_1,1} & \cdots & Y_{N+N_1,T-1} & ? & \cdots & ? \end{array}\right].$$

  Weighted regression seems to be a strong parametric assumption. Instead, Athey et al. [2021] propose to view this as a matrix completion problem, assuming an underlying low-rank structure.

2.7 When to use Synthetic Control

- Use when you have a large donor pool of control units and long pre-treatment histories.
- You are interested in the effect of a single or a small number of treated units
- Obtain individual effects, lack of external validity

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